BIOT 6900 Module 2: Multi-Omics Target Identification in Alzheimer's Disease
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1. Disease & Data Choice

I selected Alzheimer's disease (AD) for this analysis because my great grandmother passed away due to Alzheimer's. I saw the emotional impact it had on those closest to her, and I want to research this disease to understand better how to fight it. I used three independent datasets, one for transcriptomic data, one for proteomic data, and one for genomic data, to build the layers:

- Transcriptomic: RNA-seq gene expression data from NCBI GEO GSE104704 (human lateral temporal lobe, 30 samples: 8 young controls, 10 aged controls, 12 AD cases). Openly accessible, no data-use agreement required[1].
- Proteomic: TMT-based quantitative proteomics from the Emory/Seyfried lab study (ProteomeXchange PXD007160; human frontal cortex, 40 samples across Control, AD, Parkinson's, and comorbid AD/PD groups). Only the 20 samples in the Control and AD groups were used. The summarized quantification table was obtained via Synapse/AD Knowledge Portal, which required creating a free account and accepting a standard data-use agreement. This was not a significant barrier to accessing the data[2].
- Genomic: Genome-wide association data from the NHGRI-EBI GWAS Catalog, filtered to Alzheimer's disease. Openly accessible[3].

These three datasets come from entirely independent patients, so integration was done at the gene level rather than the sample level, unlike the matched CPTAC breast-cancer example in Part 1, where the same tumors were profiled at all three layers. Because no individual was profiled at all three layers, this design cannot test whether RNA and protein changes both happen within one patient, only if each is significant in their own study. This means that though any agreement between layers could be the result of biological significance, there is an increased possibility that correlation (or lack thereof) can result from differences in the patients. However, the usage of independent cohort data allows for larger datasets to be used, allowing for each layer to lead to more powerful trends.

2. Weighting Rationale

This analysis used the course-default equal weighting (1/3 each for transcriptomic, proteomic, and genomic evidence). I believe this is a reasonable split for AD. Genomics is highly important for understanding causes for potential development of AD, however it fails to provide context for individuals who actually develop the disease compared to those who don’t. Proteomics and transcriptomics provide this context, giving further understanding into how the disease manifests

and develops. Given this contextualization is so important, each of the two being equally weighted with genomic data creates a clearer picture than one that is 50% baseline genomics.

3. Top Targets

The harmonized pipeline ranked 819 gene-protein records after the merge. The top-ranked gene overall was APOE (score 0.99), followed by CLU (0.98), APP (0.96), SPEN (0.95), and MAPT (0.95). APOE, APP, and MAPT are well-established genes in AD biology: APOE is a well-known genetic risk factor for AD[4], APP is the source protein for amyloid-beta plaques and is involved a familial mutated form that corresponds with AD[5], and MAPT encodes tau, the protein of neurofibrillary tangles[4]. CLU (clusterin) has been found to be involved in amyloid clearance[6]. The ranking from this pipeline includes 4 genes known to be involved with AD is a positive sign that despite including independent layers, the datasets can present interesting correlations in a meaningful way.

Checking against a broader panel of literature-nominated AD genes: BIN1 ranked 27th (score 0.84), PICALM ranked 79th (0.74), SORL1 ranked 177th (0.65), and ABCA7 ranked 360th (0.56) out of 819[7]. TREM2 did not survive the join at all, as it was missing from the proteomics data.

The most surprising ranking in the list is SPEN. SPEN encodes SHARP, an RNA-binding protein that regulates glial cells. It is not a gene that has been widely associated with AD, however there has been some study in how the misregulation of glial cells can lead to a lessened response in protection against toxicity and neurodegeneration. Therefore, despite it not being a very well-known component of AD, studies have been conducted where SPEN is shown to be somewhat related to neurodegenerative diseases such as Alzheimer's[8].

4. A Discordant Gene: APOE

Despite ranking #1 overall, APOE shows sign-discordant evidence between layers. On one hand, rna_lfc = -2493, indicating a lower expression in AD vs. controls, but on the other, prot_lfc = +0.71, showing a higher relative abundance in AD vs. controls. Notably, the top three overall hits (APOE, CLU, APP) are all sign-discordant, which could stem partially from using the absolute value of the scores to choose the rankings.

The discordance could, however, truly stem from the biological mechanism behind the disease and the genetic pathways that lead to it. For example, APOE is involved in several mechanisms that correlate with AD. The lipidation levels of APOE can cause issues with the clearance of Aß,

and the pathology of tau, both factors in AD[9]. Given these biological pathways can be complex and varied, it follows that discordance would arise in correlated genes and proteins.

5. What This Data Lets Me Claim — and Not Claim

This analysis harmonized three independent AD cohorts at the gene level. Therefore, no sample was measured on more than one layer. This means some form of correlation between a gene or protein and AD can be claimed. For example, a gene like APOE shows independent evidence for each layer correlating somewhere in AD biology. It does not mean these three signals occur together, nor that one causes another, nor that the RNA-protein relationship reflects real biological regulation. These claims could have been made if the same patients had been used for each layer, and similar data were found.

One concrete example of the limitations of this data is the region of the brain that was inspected for each dataset. The transcriptomic data was taken from the lateral temporal lobe, whereas the proteomic data was taken from the frontal cortex, parts of the brain that serve different functions and thus could react differently to certain gene expression or protein appearance. Another limitation to note is that I used raw differences instead of logarithmic. This introduces some bias toward genes with more expression.

Despite the limitations, the data does show that there is some correlation between the top ranked genes and AD. Some of the genes present served to instill confidence in the model, such as APOE, CLU, APP, and MAPT, and some genes provided interesting possibilities into future research, such as SPEN.

References

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